

Replication Report: Hall mass and transverse Noether spin currents in noncollinear antiferromagnets (Wernert *et al.* 2024)

Automated Replication Subagent (Hermes)¹

¹*textures-100 replication campaign*
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We report a from-scratch replication of the central analytic results of Wernert, Pradenas, Tchernyshyov, and Chen, *Hall mass and transverse Noether spin currents in noncollinear antiferromagnets* (arXiv:2404.12898v2). The paper’s headline is that a *static twist* of the noncollinear kagome antiferromagnetic (AFM) order produces a purely *transverse* Noether spin current—a Hall-like response of the spin current. Using a symbolic (*sympy*) re-derivation of the continuum Noether current directly from the paper’s Lagrangian and Γ -tensor, we reproduce the headline *exactly*: $J^y = \pm(\sqrt{3}/8) JS^2 (\partial_x \phi) n_y$ with $J^x \equiv 0$ and a sign set by the direct/inverse triangular order. We further reproduce the dynamical d.c. Hall response sign structure of Fig. 2 and the polycrystal Hall-mass magnon-velocity splitting of Eq. (13). Verdict: **REPLICATED** (Coverage 8/10, Agreement 9/10).

BACKGROUND AND HEADLINE CLAIM

Noncollinear AFMs of the Mn_3X family host coplanar 120° spin order on a kagome lattice. Because global spin-rotation symmetry is fully broken by the order parameter, whether a *conserved* spin current can even be defined is nontrivial. Wernert *et al.* apply Noether’s theorem to the continuum σ -model of the kagome Heisenberg AFM and show that the conserved spin current generically acquires a component *transverse* to the gradient of the order parameter—a “Hall spin current.” The cleanest manifestation is a *static twist* of the spin frame,

$$\partial_x n_\alpha = (\partial_x \phi) n_x \times n_\alpha, \quad \partial_y n_\alpha = 0, \quad (1)$$

which the paper states yields the purely transverse current

$$J^y = \pm \frac{\sqrt{3}}{8} JS^2 (\partial_x \phi) n_y, \quad (2)$$

the $+/-$ sign corresponding to direct/inverse triangular order.

MODEL AND METHOD

The continuum Lagrangian and gradient tensor are

$$\mathcal{L} = \frac{\rho}{4} \partial_t n_\alpha \cdot \partial_t n_\alpha - \frac{1}{4} \Gamma_{ab}^{\alpha\beta} \partial_a n_\alpha \cdot \partial_b n_\beta, \quad (3)$$

$$\Gamma_{ab}^{\alpha\beta} = \eta \frac{\sqrt{3}}{4} JS^2 (\delta_{a\alpha} \delta_{b\beta} + \delta_{a\beta} \delta_{b\alpha}), \quad \alpha, \beta \in \{x, y\}, \quad (4)$$

vanishing if $\alpha = z$ or $\beta = z$; $\eta = +1$ for direct triangular order and $\eta = +1(-1)$ for $\alpha = \beta$ ($\alpha \neq \beta$) for inverse triangular order. Noether’s theorem gives the spin current

$$J^a = -\frac{1}{2} \Gamma_{ab}^{\alpha\beta} n_\alpha \times \partial_b n_\beta. \quad (5)$$

Because the paper’s core results are analytic, the faithful replication is a *symbolic* re-derivation rather

than a black-box number. We implemented Eqs. (3)–(5) in *sympy*, built Γ for both order types, and evaluated Eq. (5) for the static twist, the dynamical d.c. response $\langle J_a^\alpha \rangle = \Gamma_{ab}^{\alpha\gamma} P_b^\gamma$, and the polycrystal isotropic tensor $\bar{\Gamma}_{ab}^{\alpha\beta} = g_H(\delta_{a\alpha} \delta_{b\beta} + \delta_{a\beta} \delta_{b\alpha}) + g_0 \delta_{ab} \delta_{\alpha\beta}$. The itinerant-electron Kubo/Berry machinery of the shared `gobel2024_sd.skyrmion.kubo.Lz.kernel.py` (Göbel 2024, arXiv:2410.00820) is credited as the methodological sibling for texture-induced Hall responses; the magnonic Noether current here is derived independently.

RESULTS

T3 — Static twist (headline). Symbolic evaluation of Eq. (5) under the static twist yields $J^x = \mathbf{0}$ and $J^y = (\sqrt{3}/8) JS^2 (\partial_x \phi) \hat{e}_y$ for direct order and the negative for inverse order, matching Eq. (2) to symbolic zero, including that the current is *purely transverse*.

T4 — Dynamical d.c. Hall response. $\langle J_y^y \rangle = \Gamma_{yx}^{yx} P_x^x = \pm(\sqrt{3}/4) JS^2 P_x^x$, with the sign flipping between direct (Mn_3Ir) and inverse (Mn_3Sn) order—reproducing the opposite-sign transverse currents of Fig. 2(a) vs (b).

T5 — Polycrystal Hall mass. Diagonalizing the 3×3 dynamical matrix built from $\bar{\Gamma}$ gives one longitudinal and two degenerate transverse magnon branches whose squared-velocity splitting equals $2g_H/\rho$ exactly, confirming that the isotropic Hall mass g_H (Eq. 13, Fig. 1) controls the mode splitting and, being $O(3)$ -invariant, survives angular averaging—hence the Hall spin current is generic to noncollinear AFMs and polycrystals. We do not reproduce the exact velocity-branch *labeling* of Eq. (13) because our simplified elastic matrix omits the LLG spin-rotation projection term; only the convention-independent splitting magnitude is verified.

VERDICT AND SCORING

Verdict: REPLICATED. The headline static-twist Hall spin current is reproduced exactly, along with the dynamical sign structure and the Hall-mass mode splitting.

- **Coverage: 8/10.** All three analytic pillars covered; the full linearized-LLG FM/AFM bilayer strip numerics (Fig. 2 magnitudes, End Matter) were not rebuilt.
- **Agreement: 9/10.** T3, T4 exact; T5 splitting

exact but the Eq. (13) branch labeling not matched.

DATA AND CODE

Result JSON and the replication script are in `report/evidence/` (`wernert2024_result.json`, `wernert2024_replication.py`).
 Kubo/Berry sibling kernel credited:
`gobel2024_sd_skyrmion_kubo_Lz_kernel.py`.